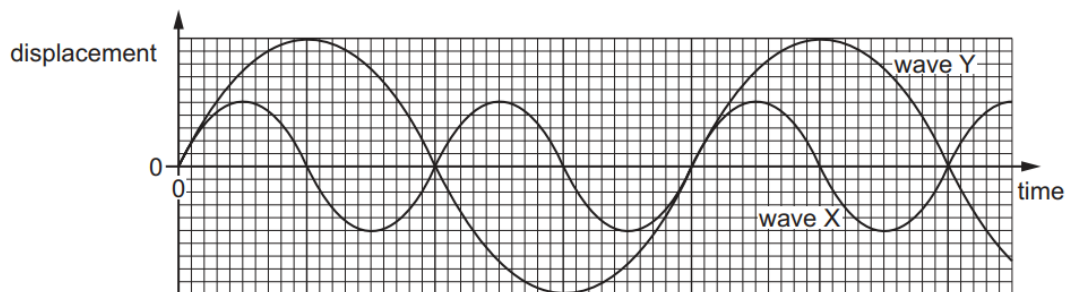


- 15** The graph shows the variation with time of displacement for two transverse waves X and Y travelling through the same medium. For such mechanical waves, intensity is proportional to the square of frequency.



Wave X has frequency f and intensity I .

What is the frequency and intensity of wave Y?

		frequency	intensity	
	A	$\frac{f}{2}$	$4I$	
	B	$\frac{f}{2}$	I	
	C	$2f$	I	
	D	$2f$	$4I$	

Answer: B

Amplitude of Wave X is A
Wave Y has amplitude = $2A$

Using the relationship $I \propto A^2 f^2$,

$$\frac{I_Y}{I_X} = \left(\frac{2A}{A}\right)^2 \times \left(\frac{\frac{f}{2}}{f}\right)^2 = 4 \times \frac{1}{4} = 1 \rightarrow I_Y = I_X = 1$$